

ASHAMitra — Modules, Backend Logic & Rules Reference

Generated reference for the clinical decision engine, the on-device pipeline, and the backend API. Engine: `asha_cdss_engine_v2`
v2.6.0-guideline-aligned (aligned to WB NHM Safe Motherhood Book 2024 + Mother & Child Protection Card 2018). Source of truth:
`assets/data/asha_engine.json` , `lib/core/services/` , `backend/server.js` .

Detailed companion docs: `MODULES.md` — every module's full rules · `PIPELINE.md` — the 11 decision steps · `SIGNOFF_RULES.md` — the 64 sign-off-pending rules. *(This file is the high-level overview + backend summary.)*

1. System Overview

ASHAMitra turns an ASHA worker's spoken/typed answers + measured vitals into a **triage band** — `GREEN` / `YELLOW` / `RED` — plus a referral, action card, and follow-up. Two cooperating parts:

- **On-device decision engine** (Dart, fully offline): a deterministic 11-layer pipeline driven by `asha_engine.json` .
- **Backend** (Node/Express + MongoDB): auth, patient records, MCH schedule + reminders, triage reports, AI chat/voice (Gemini→Groq), TTS/STT, OCR, admin.

Bands

Band	Meaning	Referral	Ambulance
 <code>GREEN</code>	Home care, routine follow-up	None	—
 <code>YELLOW</code>	Refer to PHC within 24 h	PHC within 24 h	optional
 <code>RED</code>	Refer FRU/SNCU/DH immediately	FRU / SNCU / DH	108

Core invariants

- **Evaluation order:** `hard_stop` → `combination_rules` → `numeric_rules` → `risk_score` .
- **RED is locked** — once any RED-tier rule fires, no later layer can downgrade it. The pipeline can only **escalate**, never de-escalate.
- `fallback_band`: `GREEN` — when no rule fires (no danger signs), the result is GREEN.
- **Conservative on missing data** — a missing RED-tier vital pushes `GREEN`→`YELLOW` and `YELLOW`→`RED`.
- **Deterministic + auditable** — every layer records a trace; a SHA-256 protocol hash guards the rule file.

Engine totals (v2.6.0)

7 modules · 68 questions · 42 hard-stop rules · 19 combination rules · 24 yellow rules · 42 numeric rules · **64 rules flagged** `clinical_sign_off_pending` (see §6). *(Full question text in §7 appendix.)*

2. Clinical Modules

Each module has: `questions` , `hard_stop_rules` (any one → RED), `combination_rules` (AND of answers → RED/YELLOW), `numeric_rules` (vital thresholds), `yellow_rules` (single sign → YELLOW), and a `risk_engine` (weighted score → band). All modules use score thresholds **GREEN 0–2 · YELLOW 3–5 · RED 6+** unless noted.

2.1 Newborn (0–28 days) — `newborn`

PSBI-focused. 14 questions; **every danger sign is a hard-stop RED**.

Questions: n1 can't breastfeed · n2 fever · n3 fast/difficult breathing · n4 umbilical infection · n5 lethargy · n6 yellow/blue skin · n7 convulsions · n8 hypothermia (cold to touch) · n9 skin pustules/ear pus · n10 bulging fontanelle · n11 no stool 24 h / no urine 48 h · n12 diarrhoea · n13 eye infection · n14 congenital defect.

Hard-stop → RED: NB-001...NB-008, NB-010...NB-014 (each single sign). **Yellow:** NB-Y-001 (n14 congenital defect → PHC/DH).

Combination → **RED:** fever+breathing (NB-COMB-001), jaundice+poor-feeding=kernicterus (NB-COMB-002), lethargy+breathing=sepsis (NB-COMB-003), umbilical+fever (NB-COMB-004), not-feeding+fever=PSBI (NB-COMB-005).

Numeric rules: | Rule | Vital | Threshold | Band | |---|---|---|---| | NB-VITAL-001 | SpO₂ | < 90 | RED | | NB-VITAL-002 | resp rate | ≥ 60 | RED | | NB-VITAL-003 | temp °C | ≥ 37.5 | RED | | NB-VITAL-004 | weight kg | < 1.5 | RED (VLBW) | | NB-VITAL-005 | temp °C | < 35.5 | RED (hypothermia) | | NB-VITAL-006 | heart rate | > 180 | RED | | NB-VITAL-007 | weight kg | 1.8–2.5 | YELLOW (LBW) | | NB-VITAL-008 | weight kg | < 1.8 | RED (HBNC) | | NB-VITAL-009 | temp °C | 35.5–36.4 | YELLOW (cold stress) |

Score weights: n1 3, n2 2, n3 3, n4 2, n5 3, n6 2, n7 3, n8 3, n9 2, n10 3.

2.2 Child (2 months–5 years) — **child**

IMNCI-aligned. 13 questions.

Questions: c1 fever >5 d · c2 cough/breathing · c3 diarrhoea/vomiting · c4 refusing to eat · c5 dehydration signs · c6 low weight · c7 convulsions · c8 lethargy/unconscious · c9 vomits everything · c10 stiff neck/severe headache w/ fever · c11 blood in stool · c12 pallor · c13 chest indrawing/stridor.

Hard-stop → **RED:** CH-001 (fever>5d), CH-004 (refuses feed), CH-005 (severe dehydration), CH-007 (convulsions), CH-008 (lethargy/unconscious), CH-009 (vomits everything), CH-010 (meningitis signs), CH-013 (chest indrawing/stridor). **Combination** → **RED:** breathing+dehydration (001), fever>5d+cough (002), diarrhoea+refusal (003), diarrhoea+dehydration=Plan C (004). **Yellow:** CH-002 cough, CH-003 diarrhoea, CH-006 low weight, CH-011 dysentery, CH-012 pallor.

Numeric rules: | Rule | Vital | Threshold | Band | |---|---|---|---| | CH-VITAL-001 | SpO₂ | < 90 | RED | | CH-VITAL-002 | MUAC cm | < 11.5 (SAM) | RED | | CH-VITAL-003 | MUAC cm | 11.5–12.5 (MAM) | YELLOW | | CH-VITAL-004 | resp rate | ≥ 50 (2 mo–1 y) | RED | | CH-VITAL-005a | temp °C | ≥ 40 | RED | | CH-VITAL-005b | temp °C | 39–40 | YELLOW | | CH-VITAL-006 | WFA z-score | < -3 | RED | | CH-VITAL-006b | WFA z-score | -3 to -2 | YELLOW | | CH-VITAL-007 | resp rate | ≥ 40 (1–5 y) | YELLOW | | CH-VITAL-008 | SpO₂ | 90–94 | YELLOW |

Score weights: c1 3, c2 2, c3 2, c4 3, c5 3, c6 2, c7 3, c8 3, c9 3, c10 3, c11 2, c12 2, c13 4.

2.3 Pregnancy / ANC — **pregnancy (female only)**

14 questions.

Questions: p1 high BP · p2 swelling · p3 bleeding/abdominal pain · p4 reduced fetal movement · p5 missed ANC 3+ mo · p6 blurred vision · p7 convulsion · p8 severe anaemia signs · p9 fever · p9r fever w/ rigors · p10 PROM/leaking · p11 headache · p11d severe/persistent headache · p12 dizziness.

Hard-stop → **RED:** ANC-001 (high BP), ANC-003 (APH bleeding), ANC-004 (reduced fetal movement), ANC-006 (blurred vision), ANC-007 (convulsion/eclampsia), ANC-008 (severe anaemia), ANC-009R (fever w/ rigors), ANC-010 (PROM). **Combination** → **RED:** BP+oedema (001), blurred+oedema (002), bleeding+reduced-movement (003), BP+blurred (004), headache+severe-headache (005), headache+oedema (006). **Yellow:** ANC-002 oedema, ANC-005 missed ANC, ANC-009 fever, ANC-011 headache, ANC-012 dizziness.

Numeric rules: systolic ≥140 RED (001), diastolic ≥90 RED (002), Hb <7 RED (003) / 7–10 YELLOW (004), systolic ≥160 RED (005), urine protein ≥2+ RED (006), fundal-height >4 wk diff YELLOW (007), gest. age <37 wk + labour RED (008), labour >12 h RED (009), leaking >12 h RED (010), TSH >10 RED (011) / 2.5–10 YELLOW (012), OGTT 2-h ≥140 YELLOW GDM (013), FHR <120 RED (014) / >160 RED (015).

Score weights: p1 4, p2 2, p3 4, p4 3, p5 1, p6 3, p7 4, p8 3, p9 3, p9r 4, p10 3, p11 2, p11d 2, p12 1.

2.4 Postpartum (0–42 days) — **delivery_pnc (female only)**

11 questions.

Questions: pp1 excessive bleeding/foul discharge · pp2 fever/chills · pp3 breast pain/swelling · pp4 abdominal pain/suture problem · pp5 burning urination · pp6 weakness/dizziness · pp6s fainting/collapse · pp7 convulsion · pp8 difficulty breathing · pp9 postpartum depression signs · pp10 incontinence.

Hard-stop → **RED:** PNC-001 (PPH/sepsis), PNC-006S (collapse/concealed bleed), PNC-007 (eclampsia), PNC-008 (breathing difficulty), PNC-009 (incontinence/fistula). **Combination** → **RED:** bleeding+fever=sepsis (001), abdominal-pain+fever=peritonitis (002), bleeding+weakness=shock (003), breast-pain+fever=abscess (004). **Yellow:** PNC-002 fever, PNC-003 mastitis, PNC-004 wound, PNC-005 UTI, PNC-006 anaemia, PNC-009 (pp9) depression screen.

Numeric rules: temp ≥38 RED (001), Hb <7 RED (002) / 7–10 YELLOW (003), systolic ≥140 RED (004), HR >110 YELLOW (005), ≥2 pads/30 min RED (006), >5 pads/day RED (007), temp ≥38.9 RED (008).

Score weights: pp1 4, pp2 2, pp3 1, pp4 2, pp5 1, pp6 2, pp7 4, pp8 3, pp9 1, pp6s 4.

2.5 Immunisation — immunisation (UIP schedule)

No hard-stops; all findings are YELLOW (catch-up). Questions im1 age · im2 missed vaccine · im3 days overdue · im4 currently ill · im5 booster missed · im6 prior AEFI. **Yellow:** IMM-001 missed primary, IMM-002 specific vaccine gap, IMM-003 booster, IMM-004 ill (defer), IMM-005 >3 mo overdue, IMM-006 prior AEFI (MO before next dose). **Score:** im4/im5/im6 = 1 each; thresholds GREEN 0 / YELLOW 1+. Full UIP schedule (birth → 16 y) embedded in the module.

2.6 Child Development Screening (2 mo–3 y) — development

MCP-card milestones. d1 age band · d2 any danger sign present. **Yellow:** DEV-001 (d2 → DEIC/health-centre developmental assessment). Score d2 = 3; thresholds GREEN 0 / YELLOW 1+. Milestone danger-sign lists per age band (2–3 mo ... 3 y) embedded.

2.7 Emergency — emergency (global hard-stop sweep)

Cross-cutting; **any sign** → **RED, ≤30 min to FRU/DH**. e1 bleeding · e2 convulsion/unconscious · e3 severe breathing difficulty · e4 unresponsive · e5 snakebite/animal bite · e6 poison/overdose · e7 shock · e8 major trauma/burn. Rules EM-001...EM-009. Score 4 each; thresholds GREEN 0 / RED 1+. **Also re-run against every other module's answers as a safety cross-sweep (pipeline Layer 9).**

3. On-Device Decision Pipeline (11 layers)

Orchestrated by rule_executor.dart . Runs in order; a blocking error in layers 1–5 halts with band: UNKNOWN (pipelineBlocked).

#	Layer	Job
1	Input Validator	Validates moduleId/answers/vitals types. Blocks on empty module, bad types, negative vitals.
2	Contradiction Checker	Flags impossible answers (warn) and impossible vitals (block): SpO ₂ >100, temp>45, RR>120, sys>300, dia≥sys.
3	Age/Module Validator	Age bounds per module (newborn 0–28 d, child 61 d–5 y, immunisation 0–16 y); pregnancy/PNC require female.
4	Required Vital Checker	Lists missing RED-tier vitals per module (non-blocking) → feeds conservative escalation in L8.
5	Protocol Hash Verify	SHA-256 of asha_engine.json ; mismatch within a session halts (tamper guard).
6	Rule Engine	The 4-stage evaluator (hard-stop → combination → numeric → yellow). Sets redLock , provisionalBand , triggered rules, risk score. Yellow stage skipped once RED-locked. Graded answers: "mild"/"unsure" force at least YELLOW.
7	Severity Scoring	Weighted score = answer weights + vital penalties (highest penalty per vital). Advisory band feeds L8; never downgrades RED.
8	Adaptive Risk	Escalates on demographics (newborn <7 d, LBW, infant <3 mo, pregnancy age <18/>35), history (prior RED, HRP flag, ≥2 missed ANC, RED <48 h ago), missing RED-tier vitals, and score band. Codes ADAPT_D/H/V/S_*. Never downgrades RED.
9	Safety Escalation	5 final checks: sign-off-pending flag, emergency cross-sweep , RED-lock floor enforcement, referral sanity default, GREEN-with-danger-signs → YELLOW.
10	Referral Decision	Facility + urgency + max delay per band (RED 30 min, YELLOW 24 h), transport advisory (108), travel-time est. (30 km/h), follow-up (RED-refused 4 h, YELLOW 24 h).
11	Explainable Output	Assembles DecisionOutput : band, action card (bn/en), referral, suspected conditions, danger signs, severity, adjustments, safety flags, full trace + protocol hash.

Severity vital penalties (L7): SpO₂ <90 +4 / <95 +2 · RR >60 +3 / >50 +2 · temp >40 +3 / >38.5 +2 / >37.5 +1 · sys ≥160 +4 / ≥140 +2 · dia ≥110 +3 / ≥90 +2 · Hb <7 +4 / <10 +2 · MUAC <11.5 +4 / <12.5 +2.

Supporting services

- vitals_extractor.dart — parses BP, temp (auto °F→°C if >45), MUAC, SpO₂, weight, RR, Hb from free speech; Bengali-digit normalisation; immediate spoken danger alerts.
- offline_brain.dart — picks the next most-urgent question (hard-stop 100 > combination 60 > yellow 30 > base 10, +60 if it would complete a RED combo) and fires combination/single-sign alerts; finishes at 3+ confirmed danger signs.

4. Backend API (backend/server.js)

Express + Mongoose (MongoDB Atlas). `auth` = JWT bearer; `adminOnly` = admin role.

Collections: User · Patient · Report (soft-delete) · Notification · AiCache · ScheduleEvent · Referral.

Module	Key routes
Health	<code>GET /health</code> (build marker, provider order, key count)
Auth & profile	<code>POST /api/auth/send-otp</code> , <code>POST /api/auth/verify-otp</code> (30-day JWT), <code>PUT /api/auth/profile</code> , <code>PUT /api/users/:id</code>
Patients	<code>GET/POST /api/patients</code> , <code>GET/PUT/DELETE /api/patients/:id</code> — offline-first sync (idempotent by <code>clientId</code> / <code>rchId</code> , optimistic <code>version</code> , 409 on conflict); creates schedule on save
Schedule / MCH	<code>GET /api/schedule/due</code> , <code>GET /api/schedule</code> , <code>PATCH /api/schedule/:id</code> , <code>POST /api/schedule/:id/remind</code> — ANC (4 visits from LMP), vaccines (UIP from DOB), HBNC (newborn d3–42), HBYC (child 3–15 mo). Hourly reminder scan: T-3/T-1/overdue, SMS + WhatsApp, dedup, recipient = mother
Reports	<code>POST /api/reports</code> (RED → worker "call 108" + all admins; YELLOW → worker), <code>GET/DELETE /api/reports/:id</code> , <code>PATCH /api/reports/:id/restore</code> , <code>PATCH /api/reports/repoint</code> , <code>PATCH /api/reports/:id/attach-patient</code>
Notifications	<code>GET /api/notifications</code> , <code>PATCH /:id/read</code> , <code>PATCH /read-all</code> , <code>DELETE /:id</code>
AI / triage	<code>POST /api/chat</code> , <code>POST /api/chat-with-voice</code> , <code>POST /api/detect-case</code> → shared <code>callGemini</code> (Gemini 2.5-flash primary key + rotation/benching 15 min, Groq llama-3.3-70b fallback, SHA-1 <code>AiCache</code>). Distinguishes <code>AI_QUOTA</code> vs <code>AI_FAIL</code>
Voice	<code>POST /api/tts</code> (Google Cloud bn-IN/hi-IN, tones, acronym→script rewriting, SSML pauses), <code>GET /api/voice-preview</code> , <code>POST /api/transcribe</code> (Groq Whisper, bn/hi auto-detect + Devanagari-Bengali fix)
OCR	<code>POST /api/ocr/aadhaar</code> — Secure-QR first, Tesseract fallback; returns masked Aadhaar; image discarded
Directions	<code>GET /api/directions</code> — Google route polyline for referral transport
Admin	workers CRUD + activate/deactivate, reports (filtered + deleted audit + restore + permanent), <code>GET /api/admin/stats</code> , <code>GET /api/admin/locations</code> , per-worker patients/reports/profile

Key env: `MONGO_URI` , `JWT_SECRET` , `GEMINI_API_KEY(_2/_3)` , `GEMINI_MODEL` , `GROQ_API_KEY` , `GOOGLE_TTS_API_KEY` , `GOOGLE_DIRECTIONS_API_KEY` , `SMS_*` , `WHATSAPP_*` , `USE_REAL_OTP` .

5. Triage data flow (end to end)

- ASHA speaks → STT (device or `/api/transcribe`) → `VitalsExtractor` pulls vitals.
- `offline_brain` picks next question; conversation handled by `GeminiConversationService` (online) or rules (offline).
- Answers + vitals → `RuleExecutor.execute()` → 11 layers → `DecisionOutput` (band + referral + action card).
- Result saved via `POST /api/reports` → band-based notifications.
- Patient registration (`POST /api/patients`) auto-generates the MCH schedule; the reminder engine nudges the mother before each due date.

6. Sign-off-pending rules (clinical validation needed)

64 rules carry `clinical_sign_off_pending: true` — they are live in the engine but flagged as **draft, awaiting clinician sign-off** (newer guideline additions or proxy thresholds). They still fire normally; the flag surfaces as a `SAFETY_A_001` advisory note in the output. Full list:

Newborn (12): NB-008, NB-010, NB-011, NB-COMB-002, NB-COMB-003, NB-COMB-004, NB-COMB-005, NB-VITAL-005, NB-VITAL-006, NB-VITAL-007, NB-VITAL-008, NB-VITAL-009.

Child (16): CH-004, CH-009, CH-010, CH-013, CH-COMB-002, CH-COMB-003, CH-COMB-004, CH-VITAL-004, CH-VITAL-005a, CH-VITAL-005b, CH-VITAL-006, CH-VITAL-006b, CH-VITAL-007, CH-VITAL-008, CH-011, CH-012.

Pregnancy (21): ANC-008, ANC-010, ANC-009R, ANC-COMB-003, ANC-COMB-004, ANC-COMB-005, ANC-COMB-006, ANC-VITAL-005, ANC-VITAL-006, ANC-VITAL-007, ANC-VITAL-008, ANC-VITAL-009, ANC-VITAL-010, ANC-VITAL-011, ANC-VITAL-012, ANC-VITAL-013, ANC-VITAL-014, ANC-VITAL-015, ANC-011, ANC-012, ANC-009.

Postpartum (10): PNC-001, PNC-006S, PNC-COMB-003, PNC-COMB-004, PNC-VITAL-004, PNC-VITAL-005, PNC-VITAL-006, PNC-VITAL-007, PNC-VITAL-008, PNC-009 (pp9).

Immunisation (1): IMM-006.

Emergency (4): EM-006, EM-007, EM-008, EM-009.

Note: PNC-001 (PPH hard-stop) being sign-off-pending is worth a clinician's attention — it's a core RED rule still flagged draft.

7. Appendix — All questions (every module)

Full text of every question the app can ask (68 total), grouped by module. Auto-generated from asha_engine.json v2.6.0-guideline-aligned.

Newborn Checkup (0–28 days) — newborn (14)

নবজাতক চেকআপ (০-২৮ দিন)

ID	Question (English)	প্রশ্ন (বাংলা)
n1	Is the baby unable to breastfeed?	শিশু বুকের দুধ খেতে পারছে না?
n2	Does the baby have fever?	শিশুর শরীরে জ্বর আছে?
n3	Is breathing fast or difficult?	শ্বাস-প্রশ্বাস দ্রুত বা কষ্টকর?
n4	Is the navel red, swollen, or has pus?	নাভি লাল, ফোলা বা পুঁজ আছে?
n5	Is the baby lethargic or not moving?	শিশু অনেক কম নড়ছে বা নিশ্বেজ?
n6	Does the skin look yellow or bluish?	ত্বক হলুদ বা নীলাভ দেখাচ্ছে?
n7	Has the baby had convulsions or abnormal movements?	শিশুর খিঁচুনি বা অস্বাভাবিক নড়াচড়া হয়েছে?
n8	Is the baby cold to touch / colder than normal (hypothermia)?	শিশুর শরীর ঠান্ডা লাগছে বা স্বাভাবিকের চেয়ে কম গরম?
n9	Many skin pustules / pus-filled blisters, or pus draining from ear?	ত্বকে অনেক ফুসকুড়ি/পুঁজভরা ফোঁসা, বা কান থেকে পুঁজ পড়ছে?
n10	Is the soft spot on the head (fontanelle) bulging?	মাথার নরম অংশ (তালু) ফুলে উঁচু হয়ে আছে?
n11	Has the baby NOT passed stool in the first 24 h, or NOT passed urine in the first 48 h?	শিশু কি প্রথম ২৪ ঘণ্টায় একবারও মলত্যাগ বা ৪৮ ঘণ্টায় মূত্রত্যাগ করেনি?
n12	Does the baby have loose stools / diarrhoea?	শিশুর পাতলা পায়খানা (ডায়রিয়া) হচ্ছে?
n13	Are the baby's eyes red, swollen, or draining pus?	শিশুর চোখ লাল, ফোলা বা চোখ থেকে পুঁজ পড়ছে?
n14	Does the baby have a visible congenital defect (e.g. cleft lip/palate, abnormal limb)?	শিশুর কোনো অঙ্গের জন্মগত ত্রুটি (যেমন ঠোঁট/তালু ফাটা, অস্বাভাবিক অঙ্গ) আছে?

Child Health Check (2 months–5 years) — child (13)

শিশু স্বাস্থ্য যাচাই (২ মাস-৫ বছর)

ID	Question (English)	প্রশ্ন (বাংলা)
c1	Has fever lasted more than 5 days?	পাঁচ দিনের বেশি জ্বর চলছে?
c2	Is there cough or breathing difficulty?	কাশি বা শ্বাসকষ্ট আছে?
c3	Is there diarrhoea or repeated vomiting?	ডায়রিয়া বা বারবার বমি হচ্ছে?
c4	Is the child completely refusing to eat?	শিশু খেতে একদম অস্বীকার করছে?
c5	Signs of dehydration — sunken eyes, dry lips?	পানিশূন্যতার লক্ষণ (চোখ গর্তে, ঠোঁট শুকনো)?
c6	Is weight much less than expected for age?	ওজন বয়সের তুলনায় অনেক কম?
c7	Has the child had convulsions or fits?	শিশুর খিঁচুনি হয়েছে বা হচ্ছে?

ID	Question (English)	প্রশ্ন (বাংলা)
c8	Is the child abnormally sleepy, lethargic, or unconscious?	শিশু অস্বাভাবিকভাবে ঘুমন্ত, নিস্তেজ বা অজ্ঞান?
c9	Does the child vomit everything?	যা-ই খাচ্ছে সব বমি করে ফেলছে?
c10	Stiff neck, light sensitivity, or severe headache with fever?	ঘাড় শক্ত, আলোয় কষ্ট, বা জ্বরসহ প্রচণ্ড মাথাব্যথা?
c11	Is there blood in the stool (dysentery)?	পায়খানার সাথে রক্ত যাচ্ছে?
c12	Are the palms or inner eyelids very pale (anaemia)?	হাতের তালু বা চোখের পাতা খুব ফ্যাকাশে?
c13	Lower chest indrawing, or stridor at rest?	বুকের নিচের অংশ ভেতরে ঢুকে যাচ্ছে, বা বিশ্রামেও শ্বাসের শব্দ (stridor)?

Pregnant Mother Checkup (ANC) — pregnancy (14)

গর্ভবতী মায়ের চেকআপ

ID	Question (English)	প্রশ্ন (বাংলা)
p1	Is blood pressure high?	রক্তচাপ বেশি?
p2	Is there swelling of legs or face?	পা বা মুখ ফুলে যাচ্ছে?
p3	Is there bleeding or severe abdominal pain?	রক্তপাত বা তীব্র পেট ব্যথা?
p4	Has the baby's movement reduced?	শিশুর নড়াচড়া কমে গেছে?
p5	Has ANC checkup been missed for 3+ months?	তিন মাসের বেশি ANC চেকআপ বাদ গেছে?
p6	Is there blurred vision?	চোখে ঝাপসা দেখছেন?
p7	Has there been a convulsion or fit during pregnancy?	গর্ভাবস্থায় খিঁচুনি বা ফিট হয়েছে?
p8	Very pale, weak, or breathless on slight exertion (severe anaemia)?	খুব ফ্যাকাশে, দুর্বল, বা সামান্য পরিশ্রমেই হাঁপিয়ে যাচ্ছেন?
p9	Fever, especially with chills/rigors?	জ্বর, বিশেষ করে কাঁপুনি দিয়ে জ্বর আসছে?
p10	Sudden gush or continuous leaking of fluid (PROM)?	যোনিপথে হঠাৎ জল ভেঙেছে বা ক্রমাগত জল ঝরছে?
p11	Is there a headache?	মাথা ব্যথা হচ্ছে?
p11d	Is the headache very severe, lasting > 2 days, or worsening?	মাথা ব্যথা কি খুব তীব্র, ২ দিনের বেশি, বা বাড়ছে?
p12	Is there dizziness or weakness?	মাথা ঘোরা বা দুর্বল লাগছে?
p9r	Fever WITH chills or rigors?	জ্বরের সাথে কাঁপুনি বা শীত-শীত ভাব আছে?

Postpartum Checkup (0–42 days) — delivery_pnc (11)

প্রসব পরবর্তী চেকআপ (০-৪২ দিন)

ID	Question (English)	প্রশ্ন (বাংলা)
pp1	Is there excessive bleeding or foul-smelling discharge?	অতিরিক্ত রক্তপাত বা দুর্গন্ধযুক্ত স্রাব?
pp2	Is there fever or chills?	জ্বর বা ঠান্ডা লাগছে?
pp3	Is there breast pain, swelling, or red marks?	স্তনে ব্যথা, ফোলা বা লাল দাগ?
pp4	Is there severe abdominal pain or suture problem?	পেটে তীব্র ব্যথা বা সেলাইয়ে সমস্যা?
pp5	Is there burning or difficulty urinating?	প্রস্রাবে জ্বালা বা কষ্ট হচ্ছে?
pp6	Is there extreme weakness or dizziness?	খুব দুর্বল বা মাথা ঘোরা হচ্ছে?
pp7	Has there been a convulsion or fit after delivery?	প্রসবের পর খিঁচুনি বা ফিট হয়েছে?
pp8	Is there difficulty breathing?	শ্বাস নিতে কষ্ট হচ্ছে?
pp9	Persistent low mood, crying, sleep/appetite loss, or no interest in the baby?	মন খুব খারাপ, কান্না, ঘুম/খাওয়া কমে যাওয়া, বা বাচ্চার প্রতি আগ্রহ নেই?
pp6s	Fainting / collapsing, or unable to stand (severe weakness)?	মাথা ঘুরে পড়ে যাচ্ছেন, নাকি দাঁড়াতেও পারছেন না (তীব্র দুর্বলতা)?

ID	Question (English)	প্রশ্ন (বাংলা)
pp10	Unable to control urine or stool (incontinence)?	প্রস্রাব বা পায়খানা ধরে রাখতে পারছেন না (অসংযম)?

Immunization — UIP National Schedule — immunisation (6)

টিকা মিস / ইমিউনাইজেশন

ID	Question (English)	প্রশ্ন (বাংলা)
im1	What is the child's age in months?	শিশুর বয়স কত মাস?
im2	Which vaccine was missed?	কোন টিকা মিস হয়েছে?
im3	How many days ago was the vaccine due?	কতদিন আগে টিকা দেওয়ার কথা ছিল?
im4	Does the child have any illness now?	শিশুর কি কোনো অসুস্থতা আছে এখন?
im5	Has the booster dose been missed for child aged 1–5 years?	১-৫ বছর বয়সী শিশুর বুস্টার ডোজ মিস হয়েছে?
im6	Was there a severe reaction after a previous vaccine dose (AEFI)?	আগের টিকার পর কি গুরুতর প্রতিক্রিয়া হয়েছিল?

Child Development Screening (2 months–3 years) — development (2)

শিশু বিকাশ যাচাই (২ মাস-৩ বছর)

ID	Question (English)	প্রশ্ন (বাংলা)
d1	Which age band is the child in?	শিশুর বয়স কোন ধাপে?
d2	Is ANY ONE of the danger signs listed for this age present? (check the list below)	এই বয়সের জন্য তালিকাভুক্ত কোনো একটি বিপদচিহ্ন কি দেখা যাচ্ছে? (নিচের তালিকা দেখে নিন)

Emergency — Global Hard-Stop Engine — emergency (8)

জরুরি অবস্থা যাচাই

ID	Question (English)	প্রশ্ন (বাংলা)
e1	Is there excessive bleeding?	অতিরিক্ত রক্তপাত হচ্ছে?
e2	Has there been convulsion or loss of consciousness?	খিঁচুনি বা অজ্ঞান হয়েছে?
e3	Is there severe difficulty breathing?	শ্বাস নিতে খুব কষ্ট হচ্ছে?
e4	Is the patient unresponsive or unconscious?	রোগী সাড়া দিচ্ছে না বা জ্ঞান নেই?
e5	Snakebite, venomous sting, or animal bite?	সাপে কামড়েছে, বিষাক্ত পোকা বা পশুর কামড়?
e6	Poison, pesticide, or overdose ingested?	বিষ/কীটনাশক/অতিরিক্ত ওষুধ খেয়েছে বা খাওয়ানো হয়েছে?
e7	Cold clammy skin, very weak pulse, extreme listlessness (shock)?	ঠান্ডা-ঘামে ভেজা, খুব দুর্বল নাড়ি, অত্যন্ত নিস্তেজ (শকের লক্ষণ)?
e8	Major injury, bleeding from a large wound, or severe burn?	গুরুতর আঘাত, বড় ক্ষত থেকে রক্তপাত, বা মারাত্মক পোড়া?